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Firewall Port Open & Verification (ASA GUI + Telnet)

Problem Summary

A developer required external access to an internal application server, but the connection was blocked by the firewall. I configured the required NAT and firewall rules using the Cisco ASA GUI (ASDM) to allow access to a specific port. After configuration, I verified connectivity by asking the developer to perform a telnet test to the public IP and port.

Why it matters?

- Firewall blocks inbound traffic by default
- GUI configuration does not guarantee actual connectivity

Environment

- Firewall: Cisco ASA (ASDM GUI)
- NAT Type: Static NAT (Port Forwarding)

Lessons Learned

- NAT + ACL must both be configured (one alone is insufficient)
- Telnet is a fast and simple way to validate TCP port connectivity

Troubleshooting Process

1. **Configure Firewall (GUI – ASDM)**
   - Created Static NAT rule mapping: ex) 200.100.100.42:3000 → 192.168.207.106:3000
   - Allowed traffic in Firewall Access Rules (ACL)
   - Applied rule to outside interface

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2. **Request External Validation (Telnet)**

```
telnet 200.100.100.42 3000
```

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3. **Analyze Results**

Case 1 – Success

```
Connected to 200.100.100.42
```

- Firewall + NAT + ACL working correctly

Case 2 – Connection refused

```
Connection refused
```

- Network path is open
- Application is not running on the server

Case 3 – Connection timed out

```
Connection timed out
```

- Firewall / NAT / ACL issue
- Recheck firewall configuration

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4. Internal Verification (ASA)

```
packet-tracer input outside tcp <test-ip> 12345 200.100.100.42 3000
```

- Identifies the exact drop point (ACL / NAT / routing)